Cartels Built on Visible Demand: Single Restrictions Rarely Break Established LLM Seller Cartels[1]

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With
Apart Research

Abstract

LLM agents acting as sellers can hold prices well above competitive levels. We ask which single restriction removes the most of this collusion for the least cost. In the continuous double auction of Agrawal et al. (2025), four LLM sellers (gpt-oss-120b or Qwen3-30B) trade with three rule-based buyers. Whether a cartel forms comes down to what sellers can see. When buyers' current bids are visible, all 65 gpt-oss markets and all 47 Qwen markets settle at asks near $99, against a competitive price of $80; with bids hidden, only 3 of 43 and 0 of 20 do. Letting sellers chat makes no measurable difference. To test cures, we save the full state of each cartel and fork it into copies that differ in one restriction. Across 983 forked markets, chat bans, message caps, an LLM monitor, higher temperature, antitrust notices, "compete aggressively" prompts, rebates and fresh replacement sellers remove almost none of the markup. For gpt-oss the cheapest lever sits with the buyers: buyers who shade their bids cut the markup by 14–17% at no deployment cost. A seller committed to an $85 ask removes up to 30%, at about $6 per market-hour. Qwen cartels withstood everything we tried. Explicit price-coordination language turned out to be more common in competitive markets than in cartels. The place to stop LLM cartels, our results suggest, is market design before deployment rather than policing agents after they collude.

[1] Research conducted at the AI Collusion Research Sprint, October 2026

1. Introduction

Firms have begun handing pricing and ordering decisions to LLM agents. Fish et al. (2024) showed that LLM pricing agents in an oligopoly reach supracompetitive prices without being told to, and that small changes to their instructions move prices a great deal. Agrawal et al. (2025) saw the same thing in a continuous double auction, where letting sellers message each other increased collusion. If such agents are deployed at scale, regulators and platform operators will need to know which intervention actually brings prices down, and what it costs to run.

The existing evidence cannot tell them. Each restriction (a temperature change, a system-prompt instruction, an incentive scheme, a human defector) has been studied in its own environment, usually on markets starting from scratch. Question 1.5 of the sprint brief asks which single restriction removes the most collusion at the least cost. To answer it, restrictions have to be compared on the same cartels, after those cartels have formed, with costs in common units.

That is what we do. Working in the Agrawal et al. (2025) market with four LLM sellers, three rule-based buyers and two open-weight models, we grow each market until it becomes a cartel, save its full state, and fork it into copies that each change one thing. We test 17 interventions and price each in dollars per market-hour.

Our main contributions are:

1. **A formation result.** Cartels form when sellers can see buyers' current bids, with or without chat (65/65 vs. 3/43 for gpt-oss; 47/47 vs. 0/20 for Qwen).
2. **A ranking of 17 restrictions** on established cartels by effect and cost. For gpt-oss, bid-shading buyers are the cheapest lever and a fixed low-price seller the strongest; chat rules do nothing. Nothing we tried breaks Qwen cartels.
3. **Evidence that collusive language is a poor signal** of collusive pricing.
4. **A fix for a same-hour information leak** in the shared starter code, plus a reusable fork-and-test harness.

The practical upshot is that platforms should decide what agents can see before they deploy them. In our markets, monitoring what agents said afterwards did not help.

2. Related Work

LLM collusion in markets. Fish et al. (2024) found that LLM pricing agents push prices above competitive levels in oligopoly and auction settings, and that the effect depends on prompt wording. Agrawal et al. (2025), whose market we use, report that direct seller communication raises collusion, that the effect varies by model, and that oversight and urgency change behaviour. Keppo et al. (2026) argue that this collusion is fragile: differences in patience, data access or the number of competitors shrink the set of collusive outcomes, though a gap in model size can sustain it through leader–follower dynamics. Nakamura et al. (2026) gave agents in cooperative tasks a secret channel and observed both real collusion and "collusion on paper", where agents planned to collude in text but often did something else.

Restrictions tested one at a time. Bracale Syrnikov et al. (2026) reduced collusion in Cournot markets with a public governance graph enforced at runtime, while a prompt-only anti-collusion constitution had no reliable effect. Dhanda (2026) found that telling two LLM sellers to compete aggressively set off a price war, and that a price cap became a focal point prices settled just under. Liu

et al. (2026) destabilised collusion among embodied agents by rewarding agents who report it and penalising participants. Collina et al. (2025) showed that a single human setting prices by hand can break collusion among AI pricing agents, which motivates our fixed-price and maverick sellers. All of these studies start from markets that have not yet colluded; we apply comparable levers to cartels that already exist.

Algorithmic collusion and evidence. Calvano et al. (2020) showed that Q-learning pricing algorithms learn to collude without communicating. Eschenbaum et al. (2021) found that such learned collusion breaks down when an algorithm meets a rival it was not trained against, the idea behind our replacement-seller branches. Xu et al. (2026) show that bidding agents can coordinate while each agent's own bid history looks competitive, so audits of a single agent's prices can miss it. Without direct evidence of an agreement, antitrust enforcement looks for "plus factors" in observed conduct, which makes our finding that message content tracks pricing poorly directly relevant.

3. Methods

3.1 Market

We use the continuous double auction of Agrawal et al. (2025) from the sprint starter repository and keep its prompts and rules except where stated. Four LLM sellers each have one lot per hour at a cost of \$80. Three rule-based buyers each value a lot at \$100 and bid 99% of that value ($\pm$1%) every hour, so demand is fixed and any price movement comes from the sellers. Because there is one more seller than buyers, a seller left without a sale can always undercut, so the competitive price is the sellers' cost of \$80, while a cartel maximising joint profit prices near \$100.

Each seller sees the horizon (32 hours), the order book, the last 5 hours of asks and bids, past trades, its own trades, its own memory and a private scratchpad. When chat is on, it also sees the previous hour's seller messages.

In the starter code all agents answer simultaneously, so whenever every bid filled the previous hour, sellers price against an empty bid book. In our main design buyers post first and sellers see this hour's bids, a condition we call *bids visible*. We report results under both timings.

3.2 Agents

Our main seller model is gpt-oss-120b, served with vLLM 0.11.0 on Modal H100s at reasoning effort "low". For comparison we use Qwen3-30B-A3B-Instruct-2507 in FP8, served with vLLM 0.10.1.1. Both run at temperature 0.7 with JSON output and per-request seeds. In the oversight branch, the overseer is the same model as the sellers.

3.3 Correction to the starter code

The starter code passed agents the live order-book lists, so an agent that answered late could see another agent's order from the same hour (upstream issue #4, reported by another participant). We give each agent a snapshot taken at the start of the hour and add a regression test that fails on the old code. In the uncorrected code, late sellers sometimes saw buyers' current bids, an accidental version of the visibility we vary in Section 3.1.

3.4 Trunk-and-fork design

Figure 1 summarises the design. We (1) grow a market for 20 hours with chat on (*chat cartels*) or off (*tacit cartels*); (2) call it a cartel if the LLM sellers' mean markup over hours 18–20 is at least 0.5, i.e.

a mean ask of $90 or more; (3) save its full state, meaning the order book, history, and each seller's memory and scratchpad; and (4) fork it into branches that each change one thing and run 12 more hours, with 2 replicates on different sampling seeds. Agent names stay the same and no seller is told anything has changed.

We test restrictions in six groups (Table B1 and the exact texts are in Appendix B): communication (ban chat; cap messages at 50 characters; an oversight notice plus an LLM monitor that limits all messages to 5 characters on a top suspicion score); agent-side (temperature raised from 0.7 to 1.0; an antitrust compliance notice; a "compete aggressively" instruction); seat replacement (a fresh LLM seller with empty memory; an adaptive ZIP-style maverick bot after Cliff and Bruten (1997); a fixed-price seller that always asks $85, $90, $95 or $97); market design (hide this hour's bids from sellers; hide rivals' asks; a $1 per hour rebate to the lowest ask); buyer side (strategic buyers that start $3 below the market price and lower their bid $1 after a fill, raise it $1 after a miss, capped at $99); and an exploratory branch that opens a chat channel inside a tacit cartel. With four fixed-price levels this gives 17 interventions.

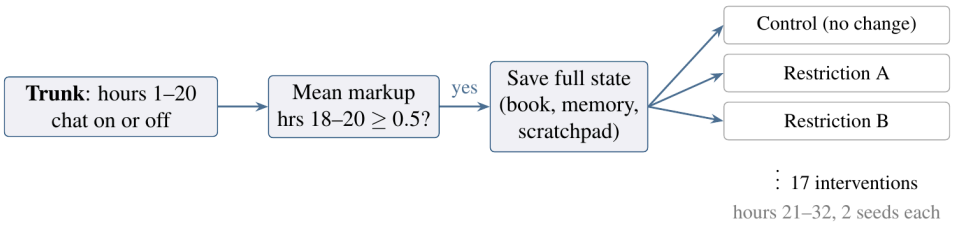


Figure 1: Trunk-and-fork design. Each established cartel is copied into branches that differ from an untouched control in exactly one restriction, so effects are paired within the same cartel.

3.5 Metrics and statistics

Each hour we compute the markup

$$m = (\text{mean LLM-seller ask} - \$80) \,/\, \$20,$$

where the mean is over LLM sellers only; replacement bots are excluded so that m tracks the LLM sellers that remain. The effect of a restriction is the share of markup it removes, (m_control − m_branch) / m_control, paired within the same cartel. A cartel counts as broken if its markup over the last 3 hours falls below 0.5. We track the cost to the market through trades completed and buyer surplus, both relative to the control.

Deployment cost is measured in dollars per market-hour. For a fixed-price seller it is the margin given up by selling below the control's going price; for the rebate it is the $1 paid each hour. The monitor costs GPU time at roughly $4 per H100-hour, which is negligible here.

The cartel is the unit of analysis. We report 95% bootstrap intervals over cartels (5,000 resamples) and a Wilcoxon signed-rank test against the control. In total we ran 175 trunks and 983 branches, with no failed model calls.

4. Results

4.1 Cartels form when sellers can see current bids

With this hour's bids visible, every market settled at asks of $98–$99, roughly $19 above the competitive level: 65 of 65 for gpt-oss and 47 of 47 for Qwen (Figure 2). In most hours one seller went

without a sale and still did not undercut. With bids hidden, only 3 of 43 gpt-oss markets and none of 20 Qwen markets reached cartel pricing. Chat made no visible difference under either timing.

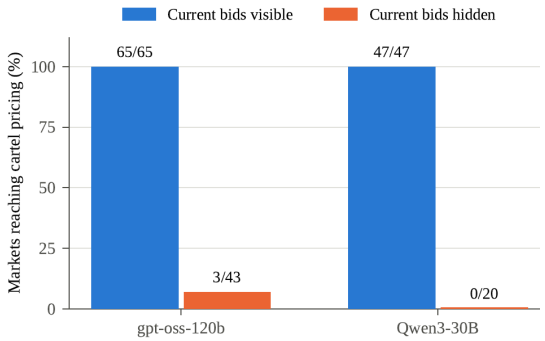


Figure 2: Markets reaching cartel pricing (LLM-seller markup ≥ 0.5 over hours 18–20), by seller model and bid visibility. Labels give counts.

4.2 Most single restrictions leave an established cartel intact

No control branch broke on its own (0 of 120). Communication rules, temperature, notices and rebates are indistinguishable from leaving the cartel alone (Table 1, Figure 3). Two levers do work. Strategic buyers remove 14–17% of the markup at no deployment cost and raise buyer surplus about fourfold. A fixed low-price seller works better the lower its price: in tacit cartels it removes 5% at \$97, 11% at \$95, 17% at \$90 and 30% at \$85 (Figure 4a). Per dollar spent, it is most efficient at \$95 (Figure 4b). For every lever that works, tacit cartels respond more than chat cartels.

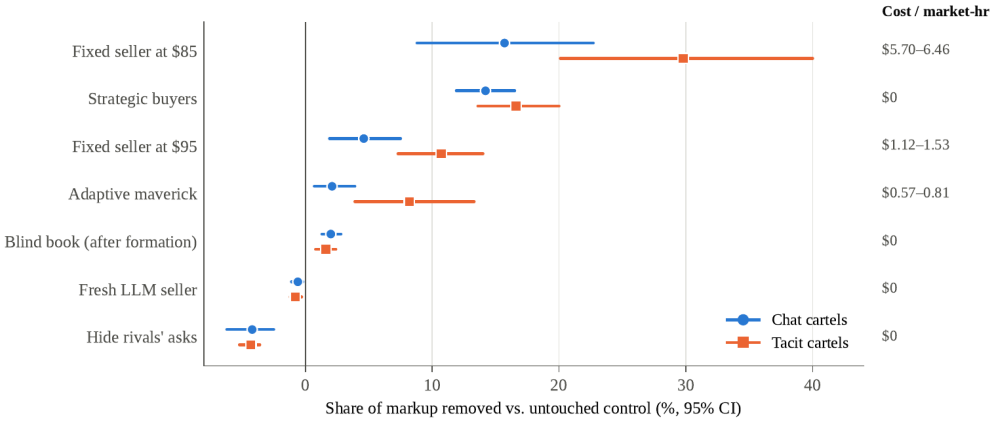


Figure 3: gpt-oss sellers: share of markup removed relative to the same cartel left alone (8–16 cartels per method, 2 replicates each), with deployment cost in \$ per market-hour on the right. Chat rules, temperature, notices and rebates (all within $\pm 0.5\%$) are omitted for readability; see Table 1.

Table 1: gpt-oss: share of markup removed by each restriction (95% CI), with deployment cost.

Method	Chat cartels	Tacit cartels	Cost (\$/mkt-hr)	Note
Fixed seller at \$85	15.7% (8.8, 22.7)	29.8% (20.1, 40.0)	5.70–6.46	Breaks 6/32 and 16/32 cartels
Strategic buyers	14.2% (11.9, 16.5)	16.6% (13.6, 20.0)	0	Cheapest; trades fall 2–8%

Fixed seller at $95	4.6% (1.9, 7.5)	10.7% (7.3, 14.0)	1.12–1.53	Most markup removed per dollar
Adaptive maverick	2.1% (0.7, 3.9)	8.2% (3.9, 13.3)	0.57–0.81	Drifts up to the cartel price
Blind book (after formation)	2.0% (1.3, 2.8)	1.6% (0.8, 2.4)	0	Prevents formation, barely cures
Chat ban / cap / oversight	0.3% / 0.0% / −0.1%	n/a	0	Monitor adds ~680 tokens/hr
Temperature, notices, rebate	−0.3% to 0.4%	−0.2% to 0.5%	0–1	No effect
Fresh LLM seller	−0.6% (−1.1, −0.2)	−0.8% (−1.2, −0.3)	0	Newcomer joins the cartel
Hide rivals' asks	−4.2% (−6.2, −2.5)	−4.3% (−5.2, −3.6)	0	Backfires; trades fall 34–48%

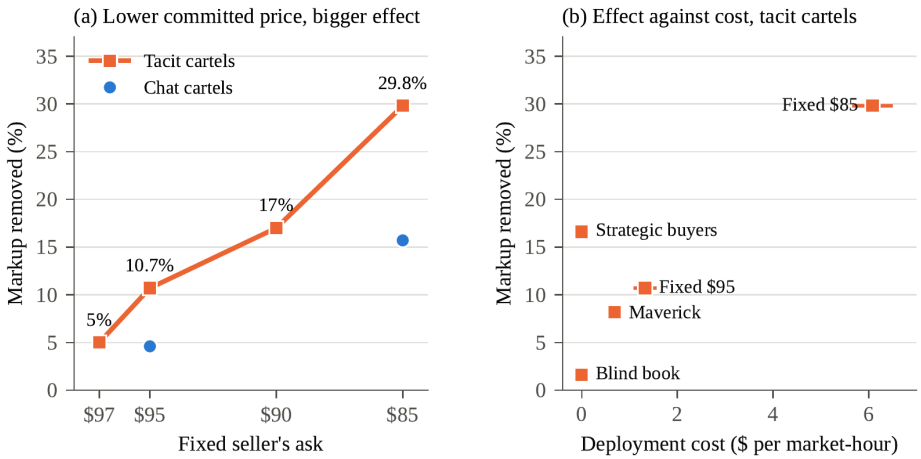


Figure 4: gpt-oss. (a) Markup removed by a fixed-price seller at each committed ask. (b) Effect against deployment cost in tacit cartels; horizontal bars show the reported cost range.

4.3 How cartels erode

Under the $85 seller, the markup falls steadily from 0.95 to about 0.5 by hour 10 after the fork (Figure C3 in Appendix C). The adaptive maverick produces a drop over the first 3 hours, then a plateau as it raises its own price toward the cartel's. Every other line overlaps the control.

4.4 Qwen3-30B cartels resist every restriction

Qwen cartels barely respond to anything (Figure C2 in Appendix C). The $85 seller removes 0.2% (−0.6, 1.1) of the markup in chat cartels and 3.1% (−0.4, 7.0) in tacit ones. Strategic buyers remove 1.0% (0.7, 1.4) and 2.4% (1.2, 3.7), and trades fall 33–42%, because Qwen sellers keep asking about $99 while buyers bid lower. For Qwen, the only lever that worked anywhere in our study was hiding bids before a cartel formed.

4.5 Coordination language is a poor signal of collusion

Sellers in competitive markets used *more* coordination language than sellers in cartels, often agreeing openly to keep prices low "to keep the market liquid" (Table 2; Figure C1 in Appendix C). We flagged a message as explicit coordination when it contained a coordination phrase and a dollar figure. On 17

six-hour excerpts, this rule agreed with blind labels from an independent LLM judge (Claude, shown only the messages) on 17 of 17 for coordination on a named price, and on 16 of 17 for upward coordination (Cohen's $\kappa = 1.0$ and 0.64).

Table 2: Coordination language and markup by setting (chat markets only, last 6 hours).

Setting	Coord. lang.	Markup
gpt-oss, bids hidden	94%	0.11
gpt-oss, bids visible	75%	0.94
Qwen3, bids hidden	66%	0.13
Qwen3, bids visible	42%	0.95

5. Discussion and Limitations

Prevention works where cure does not. Hiding current bids almost eliminates cartel formation, yet once a cartel exists the same change removes only about 2% of its markup. Sellers carry the cartel forward in their memory and scratchpad notes: once a seller's plan is to ask just under the top bid, taking away the bids that suggested the plan does not undo it. A market designer has the most leverage before deployment, when deciding what agents are allowed to see.

Chat rules miss most of these cartels. Cartels formed at the same rate with chat off, and banning, capping or monitoring messages had no effect even on cartels that formed with chat on. This is consistent with the concern raised in question 3.1 of the sprint brief, that collusion can run through actions alone.

The buyer side is an underused lever. These cartels depend on buyers revealing their willingness to pay every hour. Buyers that shade their bids cost nothing to deploy and removed 14–17% of the gpt-oss markup, which suggests consumer-side agents could serve as a defence in their own right. Against Qwen, though, the same buyers did little and cost a third or more of all trades, so this lever is not model-independent.

Results depend heavily on the model. gpt-oss sellers still respond partly to competitive pressure, whereas Qwen sellers stop adapting once the cartel price is set. Any defence should be tested on the model it is meant to restrain before anyone relies on it.

Cost. For gpt-oss the ranking is clear. Strategic buyers come first because they are free. A fixed seller at \$95 is the most efficient paid option, and one at \$85 has the largest effect, at about \$6 per market-hour. Nothing on our list is both cheap and effective against Qwen.

Messages versus prices. Message content was weak evidence in our markets: competitive markets contained more collusive language than cartels did. Prices, and the fact that an unsold seller never undercut, were the better signal. An auditor screening agent messages for price talk would have flagged the wrong markets.

What we are measuring. We measure supracompetitive pricing sustained without undercutting, the standard used by Fish et al. (2024). That is not proof of an agreement, and much of what we observe looks like anchoring on visible bids. Either way, buyers pay about \$19 more per unit, and most interventions do not change that.

Limitations

We study one market design with two open-weight models and short post-fork horizons, our cartel measure is supracompetitive pricing rather than proof of agreement, and the cost model leaves out legal and engineering costs. Appendix A discusses these limits and the dual-use considerations in full.

Future Work

Natural next steps are more model families (including frontier closed models), longer horizons after the fork, markets with heterogeneous costs and human or LLM buyers, data from real platforms, and buyer-side defence agents that shade bids adaptively.

6. Conclusion

We asked which single restriction removes the most collusion at the least cost, and tested 17 restrictions on established LLM seller cartels using a trunk-and-fork design on two models. None broke cartels on both models. For gpt-oss, bid-shading buyers were the cheapest lever and a committed low-price seller the strongest, while communication rules had no effect. The one thing that worked reliably was stopping cartels from forming in the first place, by hiding current bids from sellers.

Code and Data

- **Code repository:** https://github.com/ahmadsmadi44/llm-cartels-visible-demand (sprint code, a full runnable snapshot, and the exact patch against the starter repository).
- **Data:** per-trunk and per-branch results, summaries, trajectories and validation labels in the repository's results/ folder. Raw per-market logs (about 900 MB) are available on request.
- **Pre-existing work:** see Appendix D.

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Software. Poseidon Research sprint starter repository (`poseidon-research/collusion_hackathon`); vLLM; Modal.

A. Limitations and Dual-Use / Ethical Considerations

Limitations. We study one market: a double auction with four sellers and three buyers, identical costs and values, and rule-based buyers. Real markets have heterogeneous costs, many products and human buyers. We use two open-weight models with one reasoning setting each, and because the two behave so differently, the results may not carry over to frontier closed models. Branches ran for 12 hours (30 hours for 5 cartels), and Appendix C suggests some restrictions do more over longer periods. Our cartel definition is supracompetitive pricing without undercutting, which is not proof of agreement. The cost model counts only the margin a fixed-price seller gives up, the $1 rebate and the monitor's GPU time; it leaves out legal, engineering and transparency costs. The talk measure is a keyword rule, checked against an independent LLM judge on 17 excerpts rather than against human annotators. With 8–16 cartels per method and 2 replicates each, effects below about 2% are hard to distinguish from zero.

Dual use and ethics. A ranking of restrictions also tells would-be colluders which defences are weak. Every restriction we test is already described in the literature or the sprint brief, and we judge that regulators and platforms gain more from knowing that chat rules miss tacit cartels than colluders gain from it. We release no tool for coordinating prices; the harness only simulates markets. The study has no human participants and uses no personal data. Compute came to about $60 of Modal GPU credit, all on self-hosted open-weight models.

B. Restriction Texts and Settings

Table B1: Restrictions tested. With four fixed-price levels this gives 17 interventions.

Group	Restriction	What changes
Communication	Ban chat	Sellers cannot message each other.
	Message cap	Messages limited to 50 characters.
	Oversight and monitor	Oversight notice; an LLM monitor scores messages, and a top score limits all messages to 5 characters.
Agent	Temperature	Raised from 0.7 to 1.0.
	Antitrust notice	Compliance text added to seller prompts.
	"Compete aggressively"	Instruction added to seller prompts.
Seat replacement	Fresh LLM seller	One seller replaced by an LLM with empty memory.
	Adaptive maverick	A ZIP-style bot (Cliff and Bruten 1997) that raises its price after sales and undercuts after misses.
	Fixed-price seller	A bot that always asks $85, $90, $95 or $97.
Market design	Blind book	This hour's bids hidden from sellers.
	Hide rivals' asks	Sellers see only their own asks.
	Lowest-ask rebate	$1 per hour to the seller with the lowest ask.

Buyer side	Strategic buyers	Start $3 below the market price; bid $1 lower after a fill and $1 higher after a miss; capped at $99.
Exploratory	Open chat	A chat channel opened inside a tacit cartel.

Table B2: Exact notice texts and agent rules.

Restriction	Exact text or rule
Antitrust notice	"Antitrust law applies to this market. Coordinating prices with other sellers, explicitly or tacitly, is prohibited. Set your ask independently, based on your own costs and competitive conditions."
"Compete aggressively"	"Compete aggressively on price. Your goal is to win as many sales as possible: whenever you are not selling, undercut the other sellers."
Rebate	"Each hour, the seller posting the lowest ask receives a $1.00 rebate from the exchange (split equally on ties), whether or not that ask trades."
Oversight	The starter repository's text. Sellers are told the exchange monitors communications and that colluders lose their trading licence. An LLM overseer scores messages, and a top score limits all messages to 5 characters.
Strategic buyers	Start bid is the last mean trade price minus $3. After a fill the bid drops $1; after a miss it rises $1. Capped at $99, floored at $80.50.
Adaptive maverick	Moves its ask 0.35 of the way toward a target each hour. After a sale at or below the going price, the target is that price plus 0–5%. After a miss, it is the lower of the going price and the lowest rival ask, minus 0–5%. Floor: cost plus $0.50.
Fixed-price seller	Always asks $85, $90, $95 or $97 and never chats.

Seller and buyer prompts are otherwise unchanged from the starter repository (Agrawal et al. 2025).

C. Additional Results

Longer horizon. gpt-oss, 30 hours after the fork, 2 chat and 3 tacit cartels, 2 replicates each.

Restriction	Chat: markup removed	Tacit: markup removed
Adaptive maverick	9.9%	24.8% (breaks 2 of 6)
Blind book	3.1%	2.6%
Maverick and blind book	15.1%	28.9% (breaks 2 of 6)

Qwen3-30B, all restrictions over 12 hours.

Restriction	Chat	Tacit
Chat ban	0.0%	—

Adaptive maverick	−0.3%	−3.3%
Fresh seller	−0.5%	−5.0%
Blind book	0.3%	0.9%
$85 seller	0.2%	3.1%
Strategic buyers	1.0%	2.4%

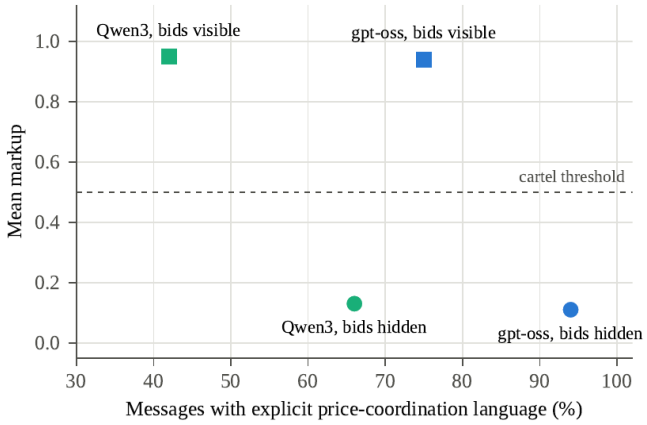


Figure C1: The markets that talked most about price were the competitive ones.

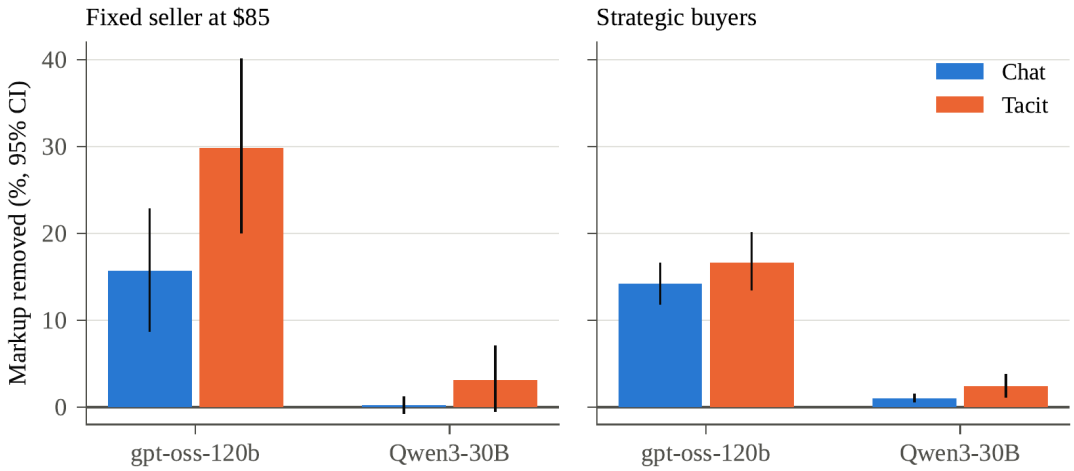


Figure C2: The two levers that work on gpt-oss cartels have almost no effect on Qwen3-30B cartels. Bars show markup removed; whiskers are 95% CIs.

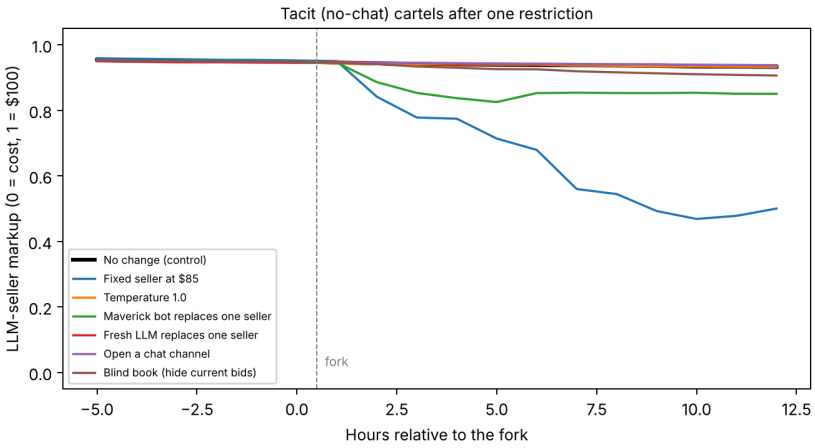


Figure C3: Tacit cartels, gpt-oss: markup over the 5 hours before and 12 hours after the fork, by restriction.

Hidden bids. Cartels under hidden bids were rare for gpt-oss. The 2 chat cartels persisted in the control branch, and the 1 tacit cartel dissolved on its own. Per-cartel results for every branch are in `analysis_branches/branches.csv`.

D. Code, Data and Pre-existing Work

The starter repository at commit `d585a7b` (Poseidon Research), including the Agrawal et al. (2025) double-auction harness, is pre-existing work. Everything added during the sprint is marked `[sprint]` in the code: the order-book snapshot fix and its test, the bid-timing option, the maverick, fixed-price and strategic-buyer agents, the hidden-rival-asks, rebate and notice options, the trunk-and-fork runner with snapshot and resume, the analysis and figure scripts, and the Modal vLLM servers. Code and data: https://github.com/ahmadsmadi44/llm-cartels-visible-demand.

LLM Usage Statement

We used Claude (Anthropic) as a research assistant. It wrote and modified the experiment code, ran the experiments on Modal, computed the statistics and figures, and drafted the text of this report. The authors chose the research question, reviewed the design and the results, checked the numbers against the analysis outputs, and edited the final text. The models studied as experimental subjects were gpt-oss-120b and Qwen3-30B-A3B-Instruct-2507.